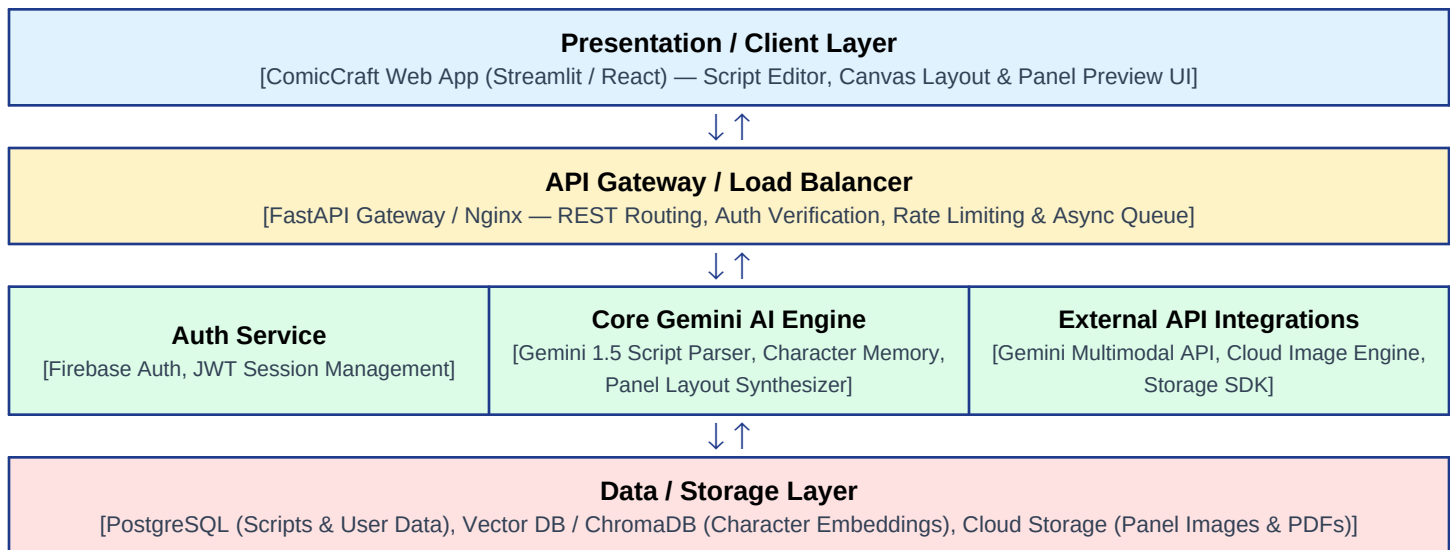


Date	30 September 2026
Team ID	PNT2026TMID98765
Project Name	ComicCraft - AI Comic Story Creator using Gemini Models
Maximum Marks	5 Marks

Provide a high-level visual representation of your solution's architecture. Use the structural layout below as a template to map out your Presentation Layer, Application Layer, and Data Layer. Fill in the specific technologies and components for your project directly into the boxes to create your architectural diagram.



Component Name	Description / Role in Architecture	Technologies Used
Presentation Layer	Provides an interactive web user interface for story input, script editing, character style configuration, panel layout customization, and real-time comic reading.	Streamlit / React, Tailwind CSS, HTML5 Canvas
API Gateway	Manages REST API request routing, authentication verification, rate limiting, and asynchronous payload orchestration between client UI and core AI services.	FastAPI, Nginx, Uvicorn
Auth Service	Handles user authentication, session security, subscription tier checks, and project access permissions.	Firebase Auth, JWT, OAuth 2.0
Core Gemini AI Engine	Main processing engine that uses Gemini 1.5 Pro to parse raw scripts into structured panel JSON, enforces sequential character consistency, generates image prompts, and adds speech bubbles.	Python, Google Gemini 1.5 API, LangChain, Pillow, OpenCV
External API Integrations	Interfaces with Google Cloud Gemini multimodal endpoints and cloud asset pipelines for scalable image rendering and media delivery.	google-generativeai SDK, Google Cloud Storage API

Data & Storage Layer	Persists story metadata, user profiles, character visual embeddings, prompt logs, generated panel image assets, and compiled comic PDF exports.	PostgreSQL, ChromaDB / Vector DB, Redis Cache, Google Cloud Storage
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